

Benjamin Pham

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EDUCATION

Virginia Commonwealth University (VCU)

Bachelor of Science: Information Systems

Richmond, VA

August 2021-May 2025

TECHNICAL SKILLS

Programming & Scripting: Python, SQL, Java, C#, HTML

Data & Analytics: Pandas, NumPy, scikit-learn, Exploratory Data Analysis, Feature Engineering, Data Modeling, Risk Scoring, Predictive Modeling

AI / Machine Learning: XGBoost, SHAP, Classification, Model Evaluation, Explainable AI, Imbalanced Learning

BI & Visualization: Tableau, Power BI, Dashboard Design, Data Storytelling, Insight Reporting

Cloud & Data Platforms: AWS, Boto3, S3, IAM, Microsoft Azure, Snowflake, Databricks, SQLite

Security & Infrastructure: Splunk Enterprise Security (SPL), CrowdStrike, Active Directory, DNS, Windows, Linux, macOS

Governance & Compliance: NIST 800-53, MITRE ATT&CK, HIPAA, SEC530

Tools: AWS CLI, Git/GitHub, ServiceNow, Archer, VirusTotal, Jira

EXPERIENCE

Virginia Department of Transportation

Security Operations Center (SOC) Analyst

Richmond, VA

December 2025-Present

- Analyze 40+ daily cases across statewide DNS, Windows, Linux, endpoint, and network datasets using Splunk Enterprise Security, independently resolving ~95% end-to-end while prioritizing high-risk issues based on operational impact.
- Query and correlate multi-source security data using SPL and SQL-based analytical techniques to validate data quality, identify anomalies, investigate behavioral patterns, and surface actionable insights across a large statewide environment.
- Develop and tune detection logic and analytics rules to improve monitoring accuracy, reduce repetitive manual analysis, and create repeatable, self-service analytical capabilities that support downstream analysts and security operations workflows.
- Translate complex technical findings into actionable insights for technical and non-technical stakeholders using structured 5W analysis, NIST, and MITRE ATT&CK methodologies while consistently meeting investigation and resolution SLAs.

Virginia Department of Health

Information Security Analyst

Richmond, VA

July 2024 - July 2025

- Evaluated system controls, security documentation, and control requirements against NIST 800-53 and SEC530 frameworks, improving data consistency, governance, traceability, and audit readiness across security documentation.
- Conducted control assessments to identify data quality and process gaps, analyzing requirements and implementation details to recommend data-driven remediation strategies in partnership with technical and business stakeholders.
- Processed and evaluated 1,000+ software installation requests against standardized security and business criteria, managing high-volume data review while maintaining accuracy, consistency, and timely processing.

PROJECTS

AI-Powered Credit Card Fraud Detection & Risk Analytics Platform

Personal Project

Richmond, VA

July 2026

- Built an end-to-end Python and SQL fraud analytics pipeline across 590K+ financial transactions, performing data cleaning, feature engineering, exploratory analysis, model development, and transaction-level risk analysis.
- Trained and evaluated an XGBoost classifier achieving 0.94 ROC-AUC, using precision, recall, F1-score, and SHAP to evaluate model performance and identify the transaction attributes driving fraud predictions.
- Applied SHAP explainability to identify the transaction attributes driving fraud predictions, translating complex machine learning outputs into interpretable insights for risk analysis and investigation prioritization.
- Developed a Low/Medium/High risk segmentation framework and quantified financial exposure from predicted fraudulent transactions, translating model outputs into actionable risk prioritization and business insights.
- Built SQL-based analytical datasets and Tableau visualizations to analyze fraud trends, transaction characteristics, model predictions, and risk distributions for business-facing reporting.

AWS Cloud Operations & Permission Monitoring

Personal Project

Richmond, VA

August 2026

- Built a Python and Boto3 automation platform to inventory AWS resources across Development, Staging, and Production environments, using AWS APIs and resource tagging to organize cloud infrastructure data.
- Automated resource discovery and security configuration checks to identify permission and access-control issues, distinguish findings by environment, and reduce repetitive manual cloud operations analysis.
- Designed a risk-based monitoring framework that evaluates AWS resource configurations, assigns severity levels to identified findings, and prioritizes issues requiring analyst attention.